

Physics — Electrostatics

Chapters 21 & 22: Electric Charge and Electric Fields

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Chapter 21 — Electric Charge

What is Electric Charge?

Electric phenomena are all around us. Walking across certain carpets and touching a metal doorknob produces a spark; pulling a sweater off can cause multiple sparks and “static cling.” These everyday effects are caused by the transfer of **electric charge** — a fundamental property of matter.

Definition: Electric Charge

Electric charge is an intrinsic characteristic of the fundamental particles making up matter. It is a property that comes automatically with those particles wherever they exist. Every object contains a vast amount of electric charge.

There exist exactly **two kinds** of electric charge:

- *Positive charge* — carried by protons in the nucleus.
- *Negative charge* — carried by electrons surrounding the nucleus.

When a body contains *equal* amounts of positive and negative charge the two contributions cancel exactly and the body is **electrically neutral** — it contains no *net* charge. If the balance is upset — for example, by rubbing two surfaces together so that electrons transfer from one to the other — then there is a net charge and the body is said to be **charged**. The charge imbalance is always far smaller than the total positive or negative charge present.

Historical note. Benjamin Franklin arbitrarily named the charge acquired by glass when rubbed with silk “positive”, and the charge acquired by rubber (or plastic) when rubbed with fur “negative”. These conventions remain in use today.

Interaction of Charges — Attraction and Repulsion

Experiment shows a simple and universal rule:

Key Rule: Attraction and Repulsion

- Two bodies with **same-sign** charge **repel** each other.
- Two bodies with **opposite-sign** charge **attract** each other.

This is the *electrostatic force*, also called the Coulomb force.

This can be demonstrated by suspending a positively charged glass rod on a pivot. Bringing a second positively charged glass rod nearby causes the suspended rod to rotate *away* (repulsion), while bringing a negatively charged plastic rod causes it to rotate *toward* (attraction).

The SI Unit of Charge

The SI unit of electric charge is the **coulomb (C)**. It is defined through the SI unit of electric current, the **ampere (A)**, via:

Electric Current and Charge (Eq. 21-3)

$$i = \frac{dq}{dt}$$

where i is current in amperes and dq is an infinitesimal charge in coulombs transferred in time dt . Rearranging:

$$dq = i dt \quad \implies \quad 1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$$

The charge of a single proton is $e = +1.602 \times 10^{-19} \text{ C}$; the charge of an electron is $-e$. A current of 1 A corresponds to roughly 6.24×10^{18} electrons passing a cross-section per second.

Conductors and Insulators

Materials differ greatly in their ability to let charge move through them.

- **Conductors** (e.g. metals): contain large numbers of *conduction electrons* that are free to move throughout the material. Any net charge placed on a conductor spreads quickly to the surface.
- **Insulators** (e.g. rubber, glass, plastic): charge cannot move freely; any charge stays essentially where it is placed.
- **Semiconductors** (e.g. silicon): intermediate behaviour; can be tuned by doping.

Grounding. Connecting a conductor to the Earth via a conducting wire (“grounding”) allows charge to flow to or from the effectively infinite reservoir of the Earth, neutralising the conductor. In the charge-sharing sample problem (Section ??), one sphere is neutralised in exactly this way.

Coulomb’s Law

The electrostatic force between two point charges was quantified experimentally by Charles-Augustin de Coulomb in 1785 using a torsion balance. His result, now called **Coulomb’s law**, is the cornerstone of electrostatics.

Coulomb’s Law — Vector Form (Eq. 21-1)

For two point charges q_1 and q_2 separated by distance r , the force exerted *on particle 1 by particle 2* is:

$$\vec{F}_{12} = k \frac{q_1 q_2}{r^2} \hat{r}_{21}$$

where $\hat{r}_{21}$ is the unit vector pointing *from* particle 2 *toward* particle 1, and $k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$.

Sign convention: The product $q_1 q_2$ is positive when the charges have the same sign, giving a repulsive (outward) force; it is negative for opposite signs, giving an attractive (inward) force.

Coulomb’s Law — Magnitude Form (Eq. 21-4)

When only the magnitude of the force is needed (direction determined by inspection):

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$$

The Coulomb Constant and Permittivity

The constant k is often written in terms of the **permittivity of free space** ϵ_0 :

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2 \quad (21-5)$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2) \quad (21-6)$$

The constant ϵ_0 appears naturally when we study electric fields in matter and in Maxwell’s equations.

Analogy with Newton’s Law of Gravitation

Coulomb’s law has the same mathematical form as Newton’s law of gravitation:

$$\vec{F}_{\text{grav}} = G \frac{m_1 m_2}{r^2} \hat{r} \quad (21-2)$$

The parallel is striking: both are inverse-square laws, both act along the line joining the two objects, and both obey superposition. However there are important differences:

	Electrostatic force	Gravitational force
Source	Charge q	Mass m
Always attractive?	No (can repel)	Yes
Relative strength	$\sim 10^{36}$ times gravity	Weakest force
Constant	$k = 8.99 \times 10^9$	$G = 6.67 \times 10^{-11}$

Superposition Principle

Superposition of Electrostatic Forces (Eq. 21-7)

When n charged particles are present, the electrostatic forces obey **superposition**: each pair of charges interacts independently of all others. The net force on particle 1 is the vector sum:

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \cdots + \vec{F}_{1n}$$

Step-by-Step Method: Net Coulomb Force

1. Identify the *particle of interest* (particle 1). Draw all other charges.
2. For each other particle j , compute $F_{1j} = \frac{1}{4\pi\epsilon_0} |q_1||q_j|/r_{1j}^2$.
3. Draw the force vector $\vec{F}_{1j}$ with its tail on particle 1. Its direction is *away* from particle j if charges are the same sign, *toward* particle j if opposite.
4. Resolve each $\vec{F}_{1j}$ into x - and y -components.
5. Sum: $F_{\text{net},x} = \sum F_{1j,x}$ and $F_{\text{net},y} = \sum F_{1j,y}$.
6. Magnitude: $F_{\text{net}} = \sqrt{F_{\text{net},x}^2 + F_{\text{net},y}^2}$. Direction: $\theta = \tan^{-1}(F_{\text{net},y}/F_{\text{net},x})$.

Worked Example 21-1: Net Force from Two Other Particles

Part (a). Two positive charges on the x -axis: $q_1 = 1.60 \times 10^{-19}$ C and $q_2 = 3.20 \times 10^{-19}$ C, separated by $R = 0.0200$ m. Find the force $\vec{F}_{12}$ on particle 1.

Both charges positive $\Rightarrow$ repulsion $\Rightarrow$ force on 1 points in $-x$ direction.

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{R^2} = (8.99 \times 10^9) \frac{(1.60 \times 10^{-19})(3.20 \times 10^{-19})}{(0.0200)^2} = 1.15 \times 10^{-24} \text{ N}$$

$$\vec{F}_{12} = \boxed{-(1.15 \times 10^{-24} \text{ N}) \hat{i}}$$

Part (b). Now add particle 3 with $q_3 = -3.20 \times 10^{-19}$ C at distance $\frac{3}{4}R = 0.0150$ m from particle 1 (between 1 and 2). Find $\vec{F}_{1,\text{net}}$.

Particle 3 is negative and closer; it attracts particle 1 in the $+x$ direction.

$$F_{13} = (8.99 \times 10^9) \frac{(1.60 \times 10^{-19})(3.20 \times 10^{-19})}{(0.0150)^2} = 2.05 \times 10^{-24} \text{ N} \Rightarrow \vec{F}_{13} = +(2.05 \times 10^{-24} \text{ N}) \hat{i}$$

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} = (-1.15 + 2.05) \times 10^{-24} \hat{i} = \boxed{(9.00 \times 10^{-25} \text{ N}) \hat{i}}$$

Part (c). Replace particle 3 with particle 4: $q_4 = -3.20 \times 10^{-19} \text{ C}$ at distance $\frac{3}{4}R$ from particle 1, on a line at $\theta = 60^\circ$ above the x -axis. Find $\vec{F}_{1,\text{net}}$.

F_{14} has the same magnitude as F_{13} above ($2.05 \times 10^{-24} \text{ N}$) because the charges and distance are identical. The direction is toward particle 4 (attraction), i.e. at 60° above the $+x$ axis. Decomposing:

$$\vec{F}_{14} = (F_{14} \cos 60^\circ) \hat{i} + (F_{14} \sin 60^\circ) \hat{j} = (1.025 \times 10^{-24}) \hat{i} + (1.775 \times 10^{-24}) \hat{j} \text{ N}$$

Adding $\vec{F}_{12} = -(1.15 \times 10^{-24}) \hat{i}$:

$$\vec{F}_{1,\text{net}} = (-1.15 + 1.025) \times 10^{-24} \hat{i} + 1.775 \times 10^{-24} \hat{j} \approx (-1.25 \times 10^{-25}) \hat{i} + (1.775 \times 10^{-24}) \hat{j} \text{ N}$$

$$F_{1,\text{net}} = \sqrt{(-1.25 \times 10^{-25})^2 + (1.775 \times 10^{-24})^2} = \boxed{1.78 \times 10^{-24} \text{ N}}$$

$$\theta = \tan^{-1} \left(\frac{1.775 \times 10^{-24}}{-1.25 \times 10^{-25}} \right) \approx -86.0^\circ$$

Worked Example 21-2: Equilibrium of Two Forces on a Particle

Two charges are fixed: $q_1 = +8q$ at the origin and $q_2 = -2q$ at $x = L$. Where on the x -axis can a proton (charge $+e$) be placed so that it is in equilibrium (zero net force)?

Step 1: Determine the region. Between the charges, $\vec{F}_1$ (repulsion from q_1) and $\vec{F}_2$ (attraction toward q_2) both point in the same direction ($+x$), so they cannot cancel. To the left of q_1 , both forces point in the $-x$ direction — again they cannot cancel. To the right of q_2 (at position $x > L$), $\vec{F}_1$ is repulsive ($+x$) and $\vec{F}_2$ is attractive ($-x$) and they can oppose each other.

Step 2: Set up force balance. Let the proton be at position x (with $x > L$). The distance to q_1 is x ; to q_2 is $x - L$:

$$F_1 = F_2 \implies \frac{1}{4\pi\epsilon_0} \frac{8q \cdot e}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{2q \cdot e}{(x - L)^2}$$

Cancel k and qe :

$$\frac{8}{x^2} = \frac{2}{(x - L)^2} \implies \left(\frac{x - L}{x} \right)^2 = \frac{2}{8} = \frac{1}{4}$$

$$\frac{x - L}{x} = \pm \frac{1}{2}$$

Taking the positive root (since $x > L > 0$ requires $x - L > 0$):

$$x - L = \frac{x}{2} \implies \frac{x}{2} = L \implies \boxed{x = 2L}$$

The proton is in **unstable** equilibrium: any slight displacement along x produces a net force that pushes it further from $x = 2L$.

Worked Example 21-3: Mutual Repulsion in a Nucleus

An iron nucleus has radius $r = 4.0 \times 10^{-15} \text{ m}$ and contains 26 protons.

(a) Find the repulsive electrostatic force between two protons separated by r .

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{(8.99 \times 10^9)(1.602 \times 10^{-19})^2}{(4.0 \times 10^{-15})^2} = \boxed{14 \text{ N}}$$

(b) Find the gravitational attraction between the same two protons ($m_p = 1.67 \times 10^{-27} \text{ kg}$).

$$F_g = G \frac{m_p^2}{r^2} = \frac{(6.67 \times 10^{-11})(1.67 \times 10^{-27})^2}{(4.0 \times 10^{-15})^2} = \boxed{1.2 \times 10^{-35} \text{ N}}$$

The ratio $F_e/F_g \approx 10^{36}$ illustrates that gravity is utterly negligible at the nuclear scale. The protons are held together not by gravity but by the strong nuclear force.

Shell Theorems for Electrostatics

By analogy with gravity, a spherically symmetric shell of charge obeys:

Shell Theorems

1. A **uniformly charged shell** exerts a force on an external charged particle as if *all* the shell's charge were concentrated at its centre.
2. A uniformly charged shell exerts **no net electrostatic force** on a charged particle located *inside* it.

These results follow from the inverse-square nature of Coulomb's law and can be proved by integration (identical in method to Newton's shell theorem for gravity).

Conservation of Charge**Law of Conservation of Charge**

The net electric charge of any *isolated* system is conserved. Charge is neither created nor destroyed; it can only be transferred between objects.

This law holds in every known physical process. Examples:

- **Pair production:** $\gamma \rightarrow e^- + e^+$ (net charge: $0 \rightarrow -1 + 1 = 0$ ✓)
- **Annihilation:** $e^- + e^+ \rightarrow \gamma + \gamma$ ($-1 + 1 \rightarrow 0 + 0$ ✓)
- **Alpha decay:** ${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He}$ ($92 = 90 + 2$ ✓)

Chapter 22 — Electric Fields

Motivation: The Electric Field

Coulomb's law describes the force between two charges — but how does charge A *know* that charge B is there? The field concept resolves this: charge A sets up an **electric field** permeating all surrounding space; charge B then responds to the field at its location. This avoids “action at a distance” and becomes indispensable in electrodynamics (changing fields propagate as electromagnetic waves at speed c).

Definition: Electric Field $\vec{E}$ (Eq. 22-1)

Place a small positive **test charge** q_0 at a point P . The electrostatic force $\vec{F}$ on it defines the electric field at P :

$$\vec{E} = \frac{\vec{F}}{q_0}$$

$\vec{E}$ is a vector with SI unit $\text{N/C} = \text{V/m}$. Its direction is the direction of force on a *positive* test charge.

The test charge must be small enough not to disturb the source charge distribution. Formally $q_0 \rightarrow 0$ is implied. Once $\vec{E}$ is known at every point, the force on any charge q placed there is simply $\vec{F} = q\vec{E}$.

Electric Field Lines

Field lines are a geometrical representation of $\vec{E}$:

- Field lines **originate** on positive charges and **terminate** on negative charges (or at infinity).
- The **tangent** to a field line at any point gives the *direction* of $\vec{E}$ there.
- The **density** (number per unit area perpendicular to the lines) represents the *magnitude* of $\vec{E}$ — closer lines mean stronger field.
- Field lines **never cross** (that would imply two directions for $\vec{E}$ at a single point).

Electric Field Due to a Point Charge

Using Coulomb's law together with the field definition:

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r} \quad \Rightarrow \quad \vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (22-3)$$

For a positive source charge $q > 0$, $\vec{E}$ points radially *outward*; for $q < 0$, it points *inward*. By superposition, the net field from n source charges is:

$$\vec{E} = \sum_{j=1}^n \vec{E}_j = \sum_{j=1}^n \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_j^2} \hat{r}_j \quad (22-4)$$

Worked Example 22-1: Net Field from Three Charges

Three charges $q_1 = +2Q$, $q_2 = -2Q$, $q_3 = -4Q$ are each at distance d from the origin. q_1 is at 150° , q_2 at -30° , and q_3 at 60° from the $+x$ axis.

Field magnitudes:

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2}, \quad E_2 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2}, \quad E_3 = \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}$$

Direction of each field: At the origin (field point), $\vec{E}_j$ points *from* the origin toward negative charges and *away* from positive charges.

$\vec{E}_1$ from $q_1 = +2Q$: points away from q_1 , i.e. at angle $150^\circ + 180^\circ = -30^\circ$.

$\vec{E}_2$ from $q_2 = -2Q$: points toward q_2 , i.e. at -30° .

So $\vec{E}_1$ and $\vec{E}_2$ are **parallel** and their sum has magnitude $\frac{1}{4\pi\epsilon_0} (4Q/d^2)$ at -30° .

$\vec{E}_3$ from $q_3 = -4Q$: points toward q_3 at 60° , magnitude $\frac{1}{4\pi\epsilon_0} (4Q/d^2)$.

Both resultants have equal magnitude; the net x -components cancel and the y -components add:

$$E = 2 E_3 \cos 30^\circ = (2) \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2} (0.866) = \boxed{\frac{6.93 Q}{4\pi\epsilon_0 d^2}}$$

The net field points in the $+x$ direction.

Charge Density

For extended (continuous) charge distributions we use charge densities:

Name	Symbol	Definition	SI Unit
Charge	q	—	C
Linear charge density	λ	$\lambda = dq/ds$	C/m
Surface charge density	σ	$\sigma = dq/dA$	C/m ²
Volume charge density	ρ	$\rho = dq/dV$	C/m ³

Full Derivation: Electric Field on the Axis of a Charged Ring

Setup. A thin ring of radius R carries total charge q uniformly distributed (linear charge density $\lambda = q/2\pi R$). We want the field at point P on the axis, at distance z from the centre.

Step 1: Contribution from one element.

Consider an infinitesimal arc element of length ds at an arbitrary location on the ring. Its charge is:

$$dq = \lambda ds$$

This element is at distance $r = \sqrt{z^2 + R^2}$ from P . By Eq. (22-3) it produces a field element of magnitude:

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{z^2 + R^2}$$

The direction of $d\vec{E}$ points from the element toward P , making an angle θ with the axis where $\cos \theta = z/\sqrt{z^2 + R^2}$.

Step 2: Cancel radial components by symmetry.

For every arc element on the ring there is a diametrically opposite element. The two $d\vec{E}$ vectors make the same angle with the axis but their *perpendicular* (radial) components point in opposite directions and cancel. Only the *axial* (z) components survive.

Step 3: Integrate axial components.

$$dE_z = dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{z^2 + R^2} \cdot \frac{z}{\sqrt{z^2 + R^2}} = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} ds$$

Since z , λ , R are all constant around the ring, integrate ds from 0 to $2\pi R$:

$$E = \int dE_z = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds = \frac{z\lambda \cdot 2\pi R}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

Since $\lambda \cdot 2\pi R = q$:

Electric Field on the Axis of a Charged Ring (Eq. 22-16)

$$E = \frac{1}{4\pi\epsilon_0} \frac{qz}{(z^2 + R^2)^{3/2}}$$

Direction: along the axis, away from (toward) the ring for positive (negative) q .

Special cases:

- At the centre ($z = 0$): $E = 0$. This confirms the symmetry argument.
- Far from the ring ($z \gg R$): $(z^2 + R^2)^{3/2} \approx z^3$, so $E \approx \frac{1}{4\pi\epsilon_0} q/z^2$ — the ring looks like a point charge.
- Maximum field: differentiate with respect to z and set to zero. This gives $z = R/\sqrt{2}$.

Full Derivation: Electric Field on the Axis of a Charged Disk

Setup. A flat disk of radius R carries surface charge density σ (C/m²). Find the field on the axis at distance z .

Step 1: Slice the disk into concentric rings.

Consider an annular ring at radius r with width dr . Its area is $dA = 2\pi r dr$ and its charge is:

$$dq = \sigma dA = \sigma \cdot 2\pi r dr$$

Step 2: Apply the ring formula.

From Section 2.5, this thin ring of charge dq contributes to the axial field:

$$dE = \frac{1}{4\pi\epsilon_0} \frac{z dq}{(z^2 + r^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{z \cdot \sigma \cdot 2\pi r dr}{(z^2 + r^2)^{3/2}} = \frac{\sigma z}{4\epsilon_0} \cdot \frac{2r dr}{(z^2 + r^2)^{3/2}}$$

Step 3: Integrate from $r = 0$ to $r = R$.

$$E = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-3/2} \cdot 2r dr$$

Use the standard integral $\int (z^2 + r^2)^{-3/2} 2r dr = -2(z^2 + r^2)^{-1/2} + C$:

$$E = \frac{\sigma z}{4\epsilon_0} \left[-\frac{2}{\sqrt{z^2 + r^2}} \right]_0^R = \frac{\sigma z}{4\epsilon_0} \left(-\frac{2}{\sqrt{z^2 + R^2}} + \frac{2}{\sqrt{z^2}} \right) = \frac{\sigma z}{2\epsilon_0} \left(\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right)$$

Electric Field on the Axis of a Charged Disk (Eq. 22-26)

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

Limiting case — Infinite Sheet ($R \rightarrow \infty$).

As $R \rightarrow \infty$ with z finite, $z/\sqrt{z^2 + R^2} \rightarrow 0$, so:

$$\boxed{E = \frac{\sigma}{2\epsilon_0}} \quad (22-27)$$

This is the field of an infinite plane of charge. Notably it is *uniform* (independent of z). The two faces of the sheet each produce $\sigma/(2\epsilon_0)$; a conducting plate held at fixed charge produces σ/ϵ_0 on one side because the interior field is zero.

Full Derivation: Electric Field of a Charged Circular Arc

Setup. A plastic rod of total charge $-Q$ is bent into a 120° arc of radius r . The axis of symmetry lies along the $+x$ axis and the origin is at the centre of curvature P . Find $\vec{E}$ at P .

Step 1: Linear charge density.

The arc subtends 120° , which is one-third of a full circle. Its arc length is:

$$s = \frac{120^\circ}{360^\circ} \cdot 2\pi r = \frac{2\pi r}{3}$$

$$\lambda = \frac{-Q}{2\pi r/3} = \frac{-3Q}{2\pi r} \approx \frac{-0.477Q}{r}$$

Step 2: Set up coordinates.

Let θ be measured from the $+x$ axis. The arc runs from $\theta = -60^\circ$ to $\theta = +60^\circ$. An element at angle θ has arc length $ds = r d\theta$ and charge $dq = \lambda r d\theta$.

Step 3: Field element.

The element is at distance r from P . Its field magnitude at P :

$$dE = \frac{1}{4\pi\epsilon_0} \frac{|dq|}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{|\lambda| r d\theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{|\lambda|}{r} d\theta$$

The direction of $d\vec{E}$ is toward the element (since $q < 0$), i.e. at angle θ from $+x$.

Step 4: Cancel y -components by symmetry.

For every element at $+\theta$ there is a symmetric element at $-\theta$; their y -components ($dE \sin \theta$ in opposite directions) cancel. Only the x -components add:

$$dE_x = dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{|\lambda|}{r} \cos \theta d\theta$$

Step 5: Integrate.

$$E = E_x = \int_{-60^\circ}^{60^\circ} \frac{1}{4\pi\epsilon_0} \frac{|\lambda|}{r} \cos \theta d\theta = \frac{|\lambda|}{4\pi\epsilon_0 r} [\sin \theta]_{-60^\circ}^{60^\circ} = \frac{|\lambda|}{4\pi\epsilon_0 r} [\sin 60^\circ - \sin(-60^\circ)]$$

$$= \frac{|\lambda|}{4\pi\epsilon_0 r} \cdot 2 \sin 60^\circ = \frac{|\lambda| \cdot 1.732}{4\pi\epsilon_0 r}$$

Step 6: Substitute $|\lambda| = 0.477Q/r$.

$$E = \frac{1.732 \times 0.477Q}{4\pi\epsilon_0 r^2} = \boxed{\frac{0.83Q}{4\pi\epsilon_0 r^2}}$$

Direction: $+x$ (toward the rod, since the rod is negatively charged).

Full Derivation: Electric Field Due to a Finite Line of Charge

Setup. A nonconducting rod of length L carries total charge $-q$ (negative) uniformly distributed. Point P is on the axis of the rod, a distance a from the nearer end. Find $\vec{E}$ at P .

Step 1: Charge density.

$$\lambda = \frac{-q}{L}$$

Step 2: Coordinate system.

Place the origin at the left end of the rod; the rod runs from $x = 0$ to $x = L$; point P is at $x = L + a$.

Step 3: Contribution from element dx .

An element of width dx at position x has charge $dq = \lambda dx$. Its distance from P is $(L + a - x)$. The element produces a field at P in the x -direction only (since P is on the axis):

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(L + a - x)^2}$$

Step 4: Integrate from $x = 0$ to $x = L$.

$$E_x = \frac{1}{4\pi\epsilon_0} \lambda \int_0^L \frac{dx}{(L + a - x)^2}$$

Let $u = L + a - x$, so $du = -dx$. When $x = 0$: $u = L + a$; when $x = L$: $u = a$:

$$\begin{aligned} E_x &= \frac{1}{4\pi\epsilon_0} \lambda \int_{L+a}^a \frac{-du}{u^2} = \frac{1}{4\pi\epsilon_0} \lambda \int_a^{L+a} \frac{du}{u^2} = \frac{1}{4\pi\epsilon_0} \lambda \left[-\frac{1}{u} \right]_a^{L+a} = \frac{1}{4\pi\epsilon_0} \lambda \left(\frac{1}{a} - \frac{1}{L+a} \right) \\ &= \frac{1}{4\pi\epsilon_0} \lambda \cdot \frac{L}{a(L+a)} \end{aligned}$$

Since $\lambda = -q/L$:

Electric Field of a Finite Charged Rod on Its Axis

$$E_x = -\frac{1}{4\pi\epsilon_0} \frac{q}{a(L+a)}$$

The negative sign means the field points in the $-x$ direction (toward the rod), which is consistent with a negatively charged rod.

Numerical values (from Sample Problem): $q = 4.23 \times 10^{-15}$ C, $L = 0.0815$ m, $a = 0.120$ m:

$$|E_x| = (8.99 \times 10^9) \frac{4.23 \times 10^{-15}}{(0.120)(0.120 + 0.0815)} = 1.57 \times 10^{-3} \text{ N/C}$$

At large distance ($a \gg L$), $L + a \approx a$, so $E_x \approx -\frac{1}{4\pi\epsilon_0} q/a^2$ (point charge). For $a = 50$ m: $|E_x| = 1.52 \times 10^{-8}$ N/C (same as a point charge).

Electric Dipole

Definition: Electric Dipole

An **electric dipole** consists of two charges $+q$ and $-q$, equal in magnitude, separated by a small distance d . The **electric dipole moment** $\vec{p}$ is a vector:

$$p = qd \quad \text{direction: from } -q \text{ to } +q$$

SI unit: C·m.

Full Derivation: Field on the Dipole Axis

Setup. Place the dipole along the z -axis, centred at the origin: $+q$ at $z = +d/2$ and $-q$ at $z = -d/2$. Find $\vec{E}$ at point P on the axis at distance z (with $z > d/2$).

Step 1: Fields from each charge.

Distance from $+q$ to P : $r_+ = z - d/2$. Distance from $-q$ to P : $r_- = z + d/2$.

$$E_{(+)} = \frac{1}{4\pi\epsilon_0} \frac{q}{(z - d/2)^2}, \quad E_{(-)} = \frac{1}{4\pi\epsilon_0} \frac{q}{(z + d/2)^2}$$

Both fields are along the $+z$ axis ($E_{(+)}$ pushes, $E_{(-)}$ pulls), but $E_{(+)} > E_{(-)}$ so the net field points in the $+z$ direction:

$$E = E_{(+)} - E_{(-)} = \frac{1}{4\pi\epsilon_0} q \left[\frac{1}{(z - d/2)^2} - \frac{1}{(z + d/2)^2} \right]$$

Step 2: Factor out z^2 .

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{1}{(1 - d/2z)^2} - \frac{1}{(1 + d/2z)^2} \right]$$

Step 3: Approximate for $z \gg d$.

Using $(1 \pm u)^{-2} \approx 1 \mp 2u$ for small $u = d/2z$:

$$\begin{aligned} \frac{1}{(1 - d/2z)^2} &\approx 1 + \frac{d}{z}, & \frac{1}{(1 + d/2z)^2} &\approx 1 - \frac{d}{z} \\ E &\approx \frac{q}{4\pi\epsilon_0 z^2} \left[\left(1 + \frac{d}{z}\right) - \left(1 - \frac{d}{z}\right) \right] = \frac{q}{4\pi\epsilon_0 z^2} \cdot \frac{2d}{z} \end{aligned}$$

Dipole Field on the Axis (Eq. 22-9)

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3} \quad (z \gg d)$$

The field falls off as $1/z^3$, faster than the $1/z^2$ of a monopole, because the two charges partially cancel.

Torque on a Dipole in an External Field

Setup. A dipole $\vec{p}$ is placed in a uniform external field $\vec{E}$, making angle θ with $\vec{E}$. The $+q$ end experiences force $+q\vec{E}$ and the $-q$ end experiences $-q\vec{E}$. The net force is zero (uniform field) but there is a net **torque** tending to align $\vec{p}$ with $\vec{E}$.

Taking torques about the dipole's centre of mass (which lies at distance x from $-q$ and $d - x$ from $+q$):

$$\tau = F \cdot x \sin \theta + F \cdot (d - x) \sin \theta = Fd \sin \theta$$

Substituting $F = qE$ and $p = qd$:

Torque on an Electric Dipole (Eq. 22-34)

$$\vec{\tau} = \vec{p} \times \vec{E} \quad \Longleftrightarrow \quad \tau = pE \sin \theta$$

Potential Energy of a Dipole

Setting the zero of potential energy at $\theta = 90^\circ$ (dipole perpendicular to $\vec{E}$), the work done by the field as the dipole rotates from 90° to angle θ is:

$$W = \int_{90^\circ}^{\theta} \tau d\theta' = \int_{90^\circ}^{\theta} (-pE \sin \theta') d\theta' = pE [\cos \theta']_{90^\circ}^{\theta} = pE \cos \theta$$

(The minus sign arises because the torque opposes increasing θ when trying to pull the dipole away from alignment.) The potential energy $U = -W$:

Potential Energy of an Electric Dipole (Eq. 22-38)

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

- Minimum ($U = -pE$) when $\theta = 0$: $\vec{p}$ parallel to $\vec{E}$ — **stable** equilibrium.
- Maximum ($U = +pE$) when $\theta = 180^\circ$: $\vec{p}$ anti-parallel to $\vec{E}$ — **unstable** equilibrium.

The work done by the external agent in rotating the dipole: $W_{\text{ext}} = \Delta U = U_f - U_i$.

Physical relevance. Water (H_2O) is a permanent electric dipole ($p = 6.2 \times 10^{-30} \text{ C}\cdot\text{m}$) because the oxygen atom pulls electron density away from the two hydrogen atoms, leaving the H-side positive and the O-side negative. This polarity underlies water's remarkable solvent properties and its anomalously high boiling point.

Motion of a Charged Particle in a Uniform Electric Field

A particle of charge q and mass m in a uniform field $\vec{E}$ experiences constant force $\vec{F} = q\vec{E}$, giving constant acceleration:

$$\vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m}$$

This is analogous to projectile motion in a gravitational field. The kinematic equations apply directly.

Worked Example 22-2: Deflection in an Ink-Jet Printer

An ink drop with mass $m = 1.3 \times 10^{-10} \text{ kg}$ and charge magnitude $Q = 1.5 \times 10^{-13} \text{ C}$ enters the space between deflecting plates of length $L = 1.6 \text{ cm} = 0.016 \text{ m}$. It moves horizontally with speed $v_x = 18 \text{ m/s}$. The field between the plates is $E = 1.4 \times 10^6 \text{ N/C}$ (directed downward, so force on the negatively charged drop is *upward*).

Step 1: Vertical acceleration.

$$a_y = \frac{QE}{m} = \frac{(1.5 \times 10^{-13})(1.4 \times 10^6)}{1.3 \times 10^{-10}} = 1.62 \times 10^3 \text{ m/s}^2$$

Step 2: Time to traverse the plates.

$$t = \frac{L}{v_x} = \frac{0.016}{18} = 8.89 \times 10^{-4} \text{ s}$$

Step 3: Vertical deflection.

$$y = \frac{1}{2}a_y t^2 = \frac{1}{2}(1.62 \times 10^3)(8.89 \times 10^{-4})^2 = 6.4 \times 10^{-4} \text{ m} = \boxed{0.64 \text{ mm}}$$

This can be written compactly as:

$$y = \frac{QEL^2}{2mv_x^2} = \frac{(1.5 \times 10^{-13})(1.4 \times 10^6)(0.016)^2}{2(1.3 \times 10^{-10})(18)^2} = 0.64 \text{ mm}$$

Summary of Key Equations

Quantity / Topic	Equation	Ref.
Electric current	$i = dq/dt$	(21-3)
Coulomb constant	$k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$	(21-5)
Permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$	(21-6)
Coulomb's law (vector)	$\vec{F} = k q_1 q_2 \hat{r}/r^2$	(21-1)
Coulomb's law (magnitude)	$F = \frac{1}{4\pi\epsilon_0} q_1 q_2 / r^2$	(21-4)
Superposition	$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \dots$	(21-7)
Electric field definition	$\vec{E} = \vec{F}/q_0$	(22-1)
Field of a point charge	$\vec{E} = \frac{1}{4\pi\epsilon_0} q \hat{r}/r^2$	(22-3)
Field on axis of ring	$E = \frac{1}{4\pi\epsilon_0} qz/(z^2 + R^2)^{3/2}$	(22-16)
Field on axis of disk	$E = (\sigma/2\epsilon_0) \left(1 - z/\sqrt{z^2 + R^2}\right)$	(22-26)
Infinite sheet	$E = \sigma/(2\epsilon_0)$	(22-27)
Field of finite rod (axis)	$E_x = -\frac{1}{4\pi\epsilon_0} q/[a(L+a)]$	—
Dipole moment	$p = qd$ (from $-q$ to $+q$)	—
Dipole axial field ($z \gg d$)	$E = p/(2\pi\epsilon_0 z^3)$	(22-9)
Torque on dipole	$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta$	(22-34)
Dipole potential energy	$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$	(22-38)
Particle deflection	$y = QEL^2/(2mv_x^2)$	—

Key conceptual comparisons to keep in mind:

- Coulomb's law is formally identical to Newton's gravity but $\sim 10^{36}$ times stronger and can be repulsive.
- The electric field $\vec{E}$ is the force per unit charge; it exists even when no test charge is present.
- Field lines originate on positive charges and terminate on negative charges; their density encodes field strength.
- For continuous distributions, always exploit symmetry to determine which components survive before integrating.
- The $1/r^3$ dipole field decays faster than the $1/r^2$ point-charge field because the two charges partially cancel at large distances.